

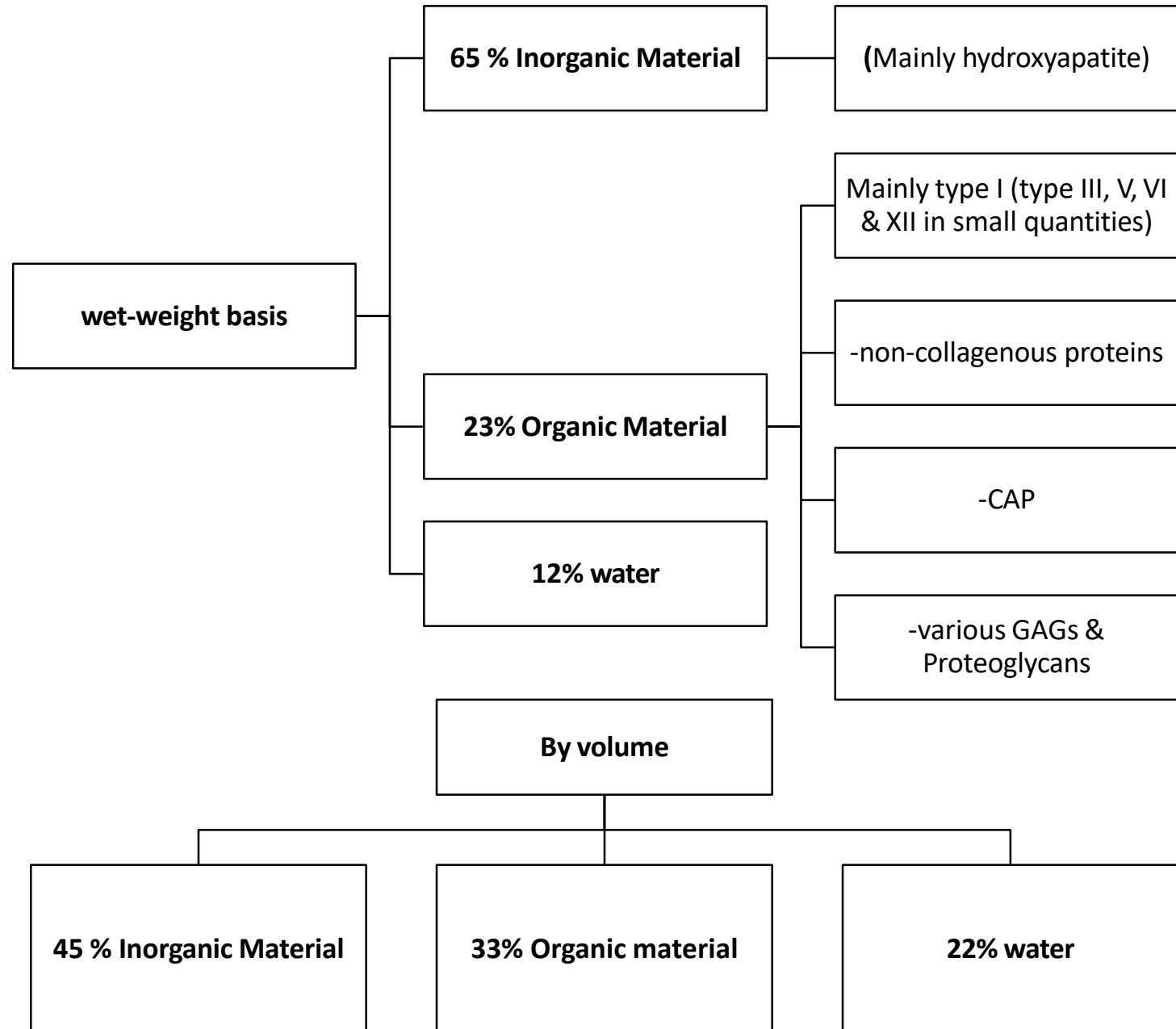
CEMENTUM (TUTORIAL)

DR. MISBAH ASHRAF MOTEN



TASK 1: ANSWER THE FOLLOWING QUESTIONS

Q1) What is the composition of cementum?



Q2) Classify different types of cementum?

Based on the nature & origin of organic matrix

Intrinsic fibres Cementum → (fibres derived from cementoblasts) parallel to the root surface, perpendicular to extrinsic fibres

Extrinsic Fibres Cementum → (from PDL as Sharpey's Fibers) they are either perpendicular or oblique to the root surface

Mixed fibre cementum → (both extrinsic & intrinsic fibres present)

Based on presence or absence of cells

Acellular (Primary Cementum) → Slower forming, relatively structureless, first formed cementum without cells, covers root dentin

Cellular (Secondary Cementum) → Faster forming variety having cementocytes, covers apical & interradicular areas and areas overlying acellular cementum

Based on absence of cells & nature & origin of fibres (widely used)

Acellular Extrinsic Fire Cementum (AEFC) → First formed Cementum, constitutes bulk of cementum in some teeth, located mainly over the cervical 2/3rds of the root

Cellular Intrinsic Fibre Cementum (CIFIC) → found in apical 3rds and inter-radicular areas in patches, no role in tooth attachment, cementoid seam

Acellular intrinsic Fibre Cementum → Slow formation rate with cementoid seam

Afibrillar Cementum → Acellular, sparsely distributed, covers cervical enamel, intervenes b/w fibrillar cementum dentin

Mixed fibre Cementum → derived from both extrinsic and intrinsic fibres

Acellular Mixed Fibre Cementum → (when rate of formation is slow, well mineralized)

Cellular Mixed Fibre Cementum → (when rate of formation is fast, less well mineralized)

Q3. Mention the differences between cellular and acellular cementum?

Table 11.1 Summary of differences between acellular and cellular cementum

Acellular cementum	Cellular cementum
No cells	Lacunae and canaliculi containing cementocytes and their processes
Border with dentine not clearly demarcated	Border with dentine clearly demarcated
Rate of development relatively slow	Rate of development relatively fast
Incremental lines relatively close together	Incremental lines relatively wide apart
Precementum layer virtually absent	Precementum layer present

Q4. How is A fibrillar Cementum different with rest of cementum?

1. Lacks collagen fibrils, unlike other types (intrinsic, extrinsic, mixed).
2. Thin, acellular, and well-mineralized.
3. Covers cervical enamel or lies between fibrillar cementum and dentin.
4. Likely forms after loss of reduced enamel epithelium.
5. Possibly of epithelial origin.

Q5. What are the different arrangements of Cemento-Enamel Junction (AKA OMG RULE)?

- CEMENTOENAMEL JUNCTION (CEJ) VARIATIONS:
 - 60% Cementum overlaps enamel
 - 30% Edge to edge
 - 10% Dentin exposed
- Cementodentinal Junction (CDJ)
- **Clinical Note:** Hypersensitivity due to
 - exposed dentin.

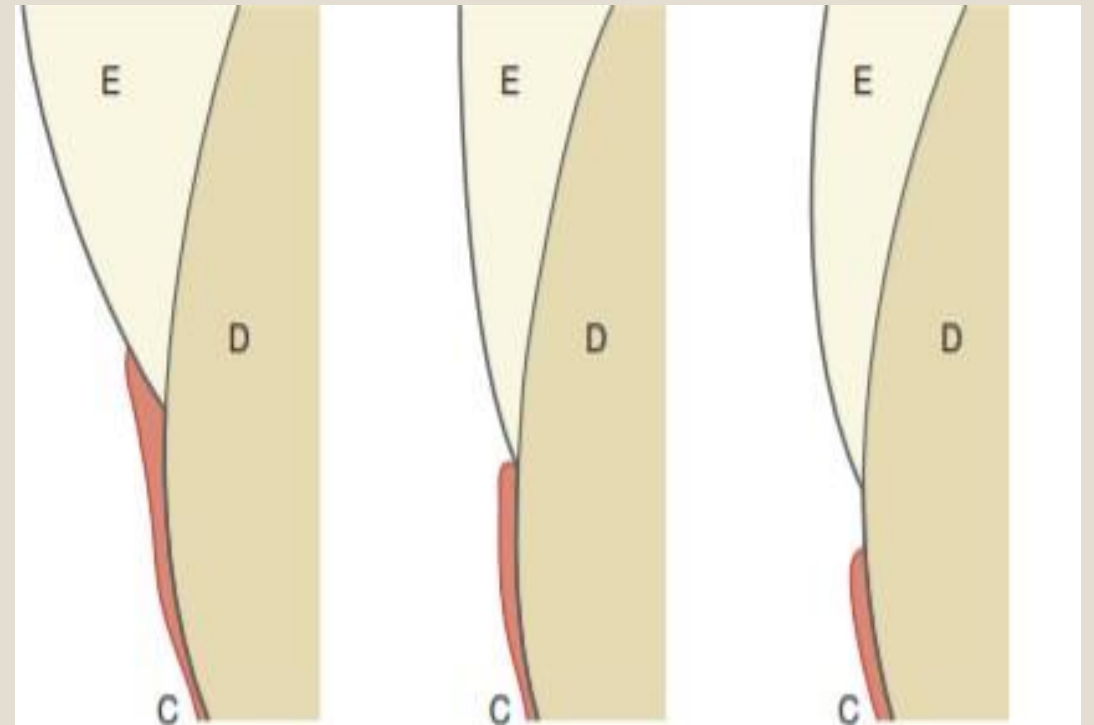


Fig. 11.3 Three patterns for the arrangement of the cementum-enamel junction. C = cementum; D = dentine; E = enamel. See text for further details.



FIGURES TO BE
PASTED IN JOURNAL

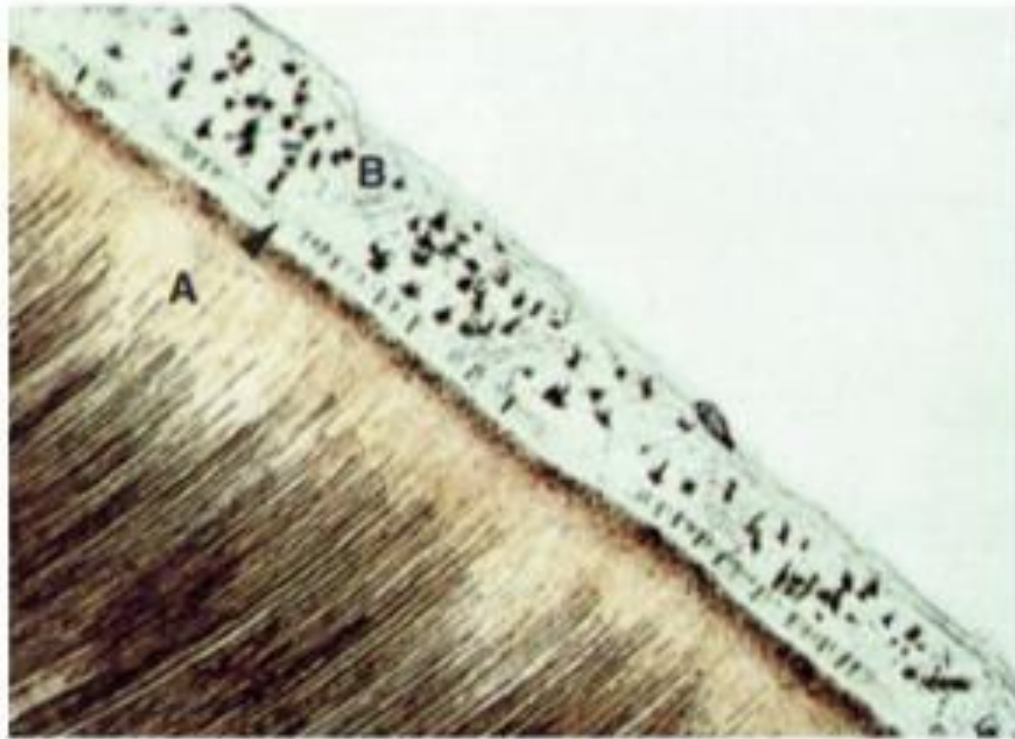


Fig. 11.8 Cellular cementum (B) overlying acellular cementum (A). Note the greater thickness of the cellular layer (ground section; $\times 50$).

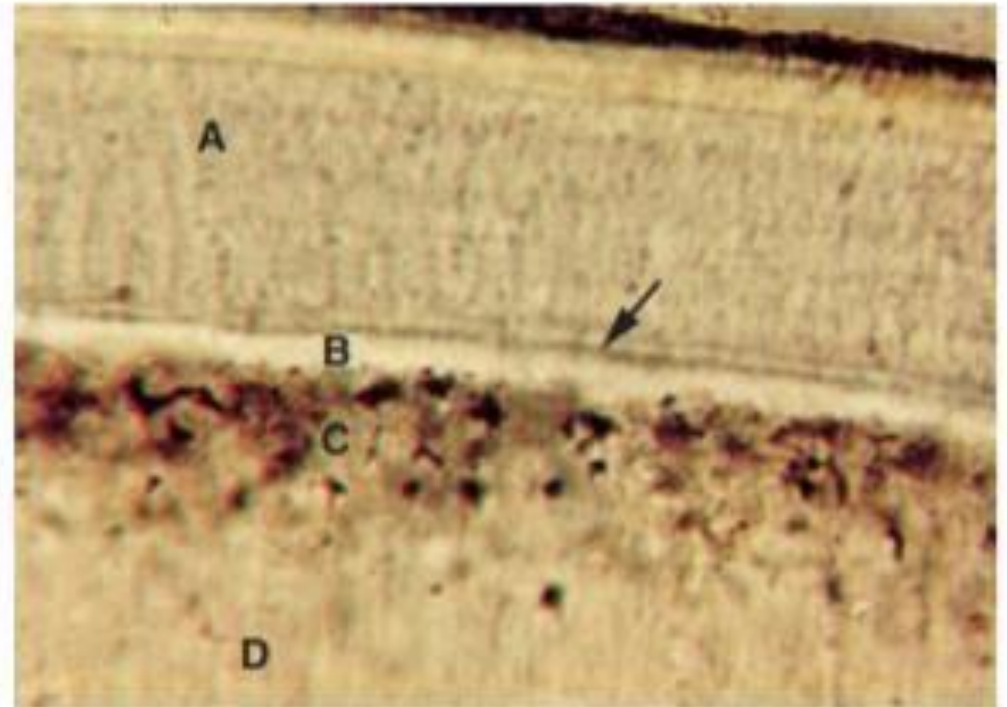


Fig. 11.7 The appearance of acellular cementum (A). B = hyaline layer (of Hopewell-Smith); C = granular layer (of Tomes); D = root dentine. Note that the dark layer arrowed between the hyaline layer and the acellular cementum may be related to the afibrillar cementum patchily present at this position (ground section; $\times 200$).

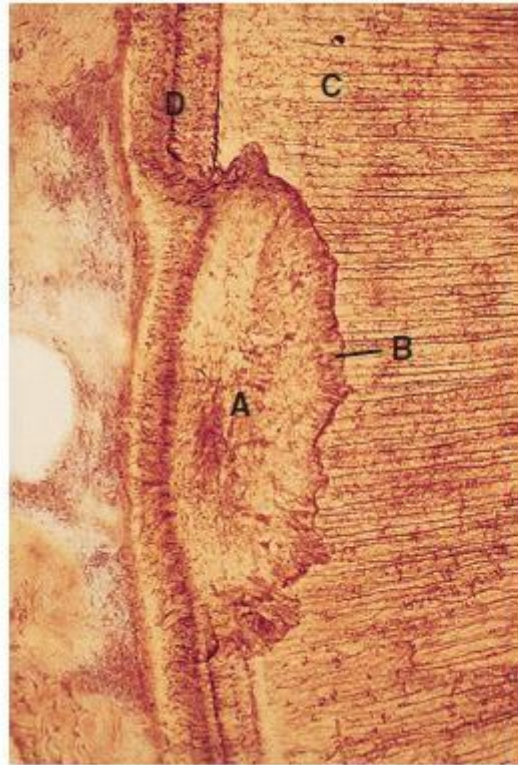


Fig. 11.29 Demineralised section of root showing infilled area of repair cementum (A). B = reversal line; C = dentine; D = cementum (picrothionin; $\times 75$). Courtesy Prof. M M Smith.

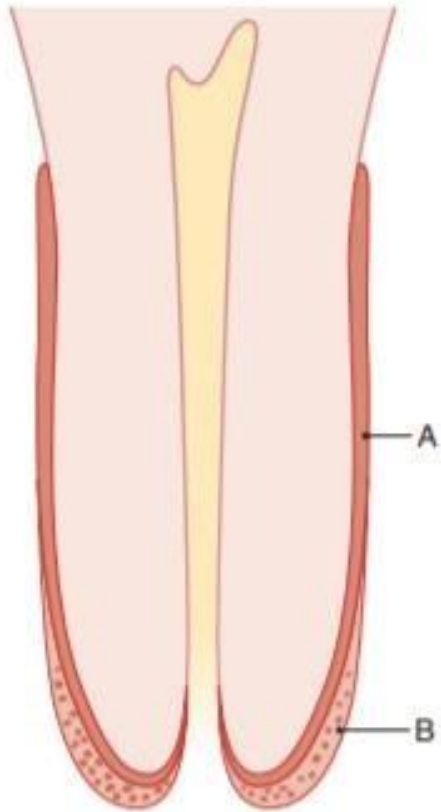


Fig. 11.6 The distribution of acellular (A) and cellular (B) cementum.

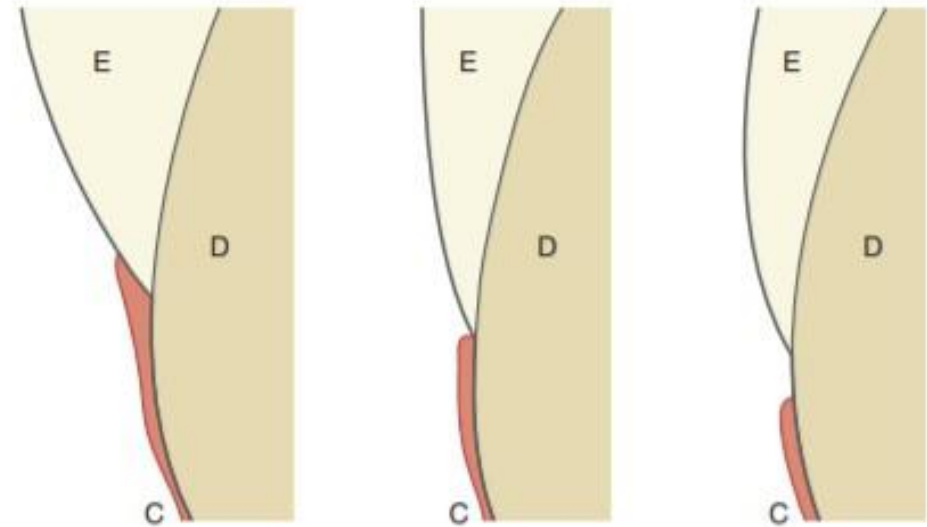
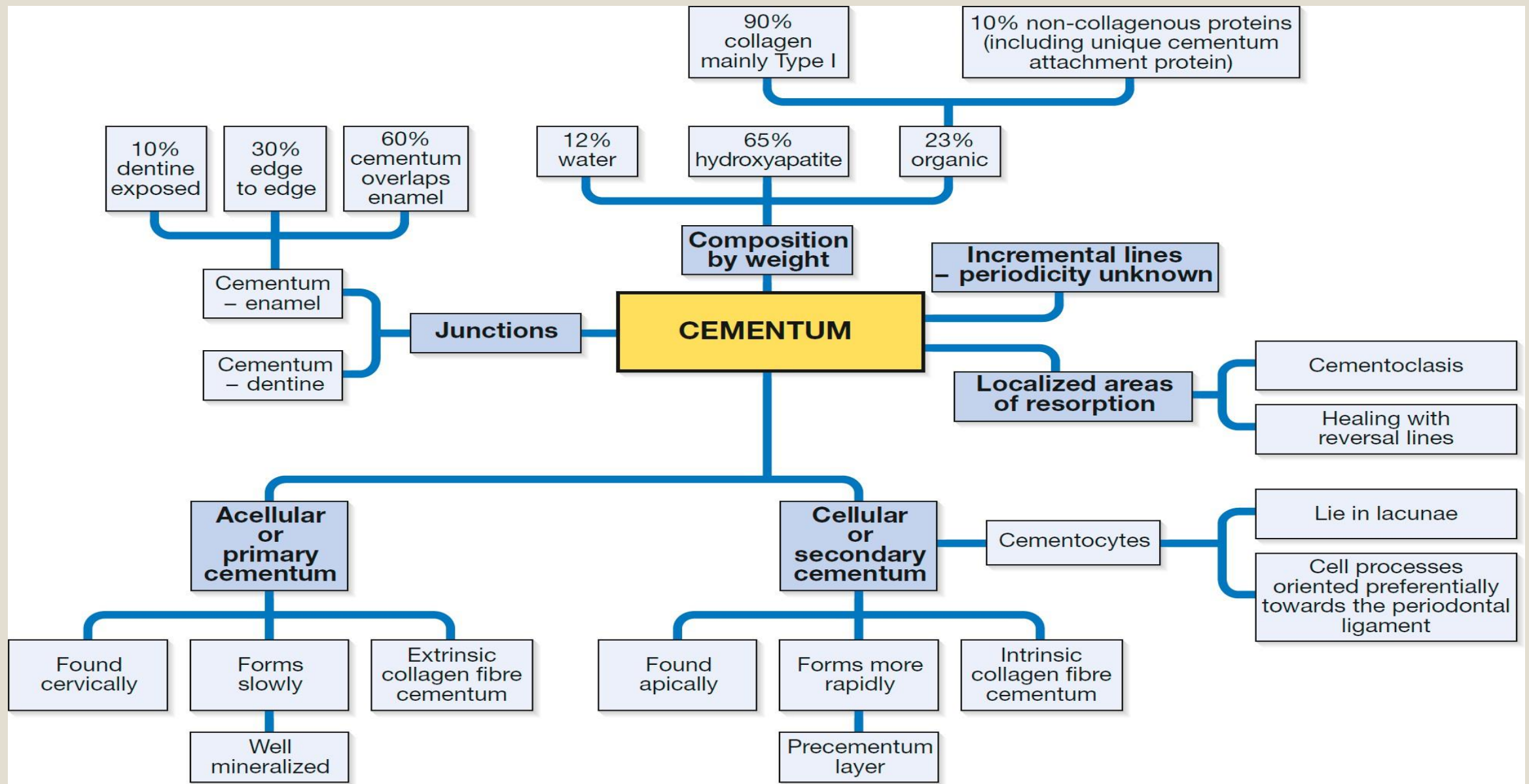


Fig. 11.3 Three patterns for the arrangement of the cementum-enamel junction. C = cementum; D = dentine; E = enamel. See text for further details.



READING MATERIAL



What is Cementoclasia?

1. **Cementoclasia** refers to resorption of cementum, often due to trauma, pressure (e.g. excessive orthodontic forces), or pathology.
2. **Healing Response:**
 - Cementum has the ability to repair itself.
 - Repair occurs by deposition of new cementum in the resorbed areas.
 - These repaired zones are separated by reversal lines—they indicate past resorption and new deposition.

KEY SIMILARITIES BETWEEN CEMENTUM AND ALVEOLAR BONE

Feature	Cementum	Alveolar Bone
Nature	Mineralized connective tissue	Mineralized connective tissue
Composition	Contains mineral, collagen and non-collagenous proteins	Contains mineral, collagen and non-collagenous proteins
PDL Attachment	Provides attachment for PDL fibers	Provides attachment for PDL fibers
Collagen Fibers	Contains collagen and Sharpey's fibers	Contains collagen and Sharpey's fibers
Forming Cells	Formed by cementoblasts	Formed by osteoblasts
Mature Cells	Contains cementocytes in lacunae	Contains osteocytes in lacunae
Functional Adaptation	Responds to functional forces	Responds to functional forces
Lifelong Activity	Continues to be deposited throughout life	Continuously formed and remodeled throughout life

Why is cementum less readily resorbed than bone?

- **Precementum/collagen layer** – The unmineralized collagen-rich layer covering cementum makes the mineralized surface less accessible to odontoclasts.
- **Dense Sharpey's fibers** – Particularly in **acellular cementum**, the greater density of Sharpey's fibers may increase resistance to resorption.
- **Cementoblasts remain at the surface** – Cementoblasts may not retract from the root surface in response to parathormone, thereby preventing exposure of the cementum surface to odontoclasts.
- **Higher fluoride content** – Cementum contains more fluoride than bone, which may contribute to its greater resistance to resorption.
- **Different physicochemical properties** – Differences in the biological and physicochemical properties of cementum compared with bone may also contribute to its resistance.

KEY DIFFERENCES BETWEEN CEMENTUM AND BONE

Feature	Cementum	Alveolar Bone
Location	Covers the root dentin	Forms the bony socket surrounding the tooth
Main function	Provides attachment of PDL fibers to the tooth and helps maintain root integrity	Supports the tooth and provides attachment of PDL fibers to the bone
Forming cell	Cementoblasts deposit cementum matrix	Osteoblasts deposit bone matrix
Mature cell	Cementocytes lie in lacunae within cellular cementum	Osteocytes lie in lacunae within bone
Vascularity	Avascular ; receives nutrients mainly by diffusion from the PDL	Vascular ; contains blood vessels within its bone tissue
Innervation	No direct nerve supply	Innervated and associated with sensory nerve fibers

Feature

Location

Cementum

Covers the **root dentin**

Remodeling

Slow/minimal remodeling;
continues to be deposited
throughout life

Resorption

Relatively resistant to resorption,
helping protect the tooth root
during orthodontic movement

Surface matrix

Covered by
precementum/cementoid, an
unmineralized layer that protects
the surface

Thickness

Thin: ~10–15 μm cervically and 50–
200 μm apically; may become
thicker with age

Alveolar Bone

Forms the **bony socket** surrounding the tooth

Continuous remodeling through coordinated
bone formation and resorption

More readily resorbed and remodeled in
response to mechanical forces

Has an **osteoid** layer on actively forming bone
surfaces

Variable according to the location and
functional demands of the tooth-supporting
socket

Mineralization of Cementum

- **Cementum mineralization begins in the precementum (cementoid)** adjacent to the root dentin.
- Mineralization is probably **initiated by the underlying mineralized root dentin** and progresses outward.
- Mineral deposition occurs **along and around collagen fibers** produced by cementoblasts and the **Sharpey's fibers** incorporated from the PDL.
- The outer cementum, where **Sharpey's fibers are abundant**, may therefore be regarded as a **calcified component of the periodontal ligament**.
- Unlike dentin, **calcospherites are absent** during cementum mineralization.
- Mineralization is **ongoing throughout life**, contributing to the gradual increase in cementum thickness.
- Near the **cementum–dentin junction**, acellular cementum may resemble peripheral dentin, making the boundary difficult to distinguish.

Q5. What is cements-dentine junction and its importance?

1. Interface between cementum and dentin with lower mineral content.
2. Contains organic/inorganic components (e.g., collagen, dermatan sulphate).
3. May include an intermediate layer anchoring PDL fibres and acting as a barrier.
4. Supports cementogenesis and healing.
5. Important for periodontal regeneration after surgery.